

Operating Systems Structures

DSC 315: Computer Organization & Operating Systems

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1 Operating Systems Structures

- Operating System Services
- System Calls
- API – System Call – OS Relationship
- System Services
- The Role of the Linker and Loader
- Why Applications are Operating System Specific?

Operating Systems Structures

Environment for executing programs and supporting users

■ User Services

- **User Interface** – interaction with system (CLI, GUI, touch)
- **Program Execution** – load, run, and terminate programs
- **I/O Operations** – read and write to devices or files
- **File-System Manipulation** – create, delete, read, write files
- **Communication** – exchange data between processes
- **Error Detection** – detect and handle system errors

■ System Services

- **Resource Allocation** – manage CPU, memory, and devices
- **Logging** – record usage of system resources
- **Protection and Security** – control access and authenticate users

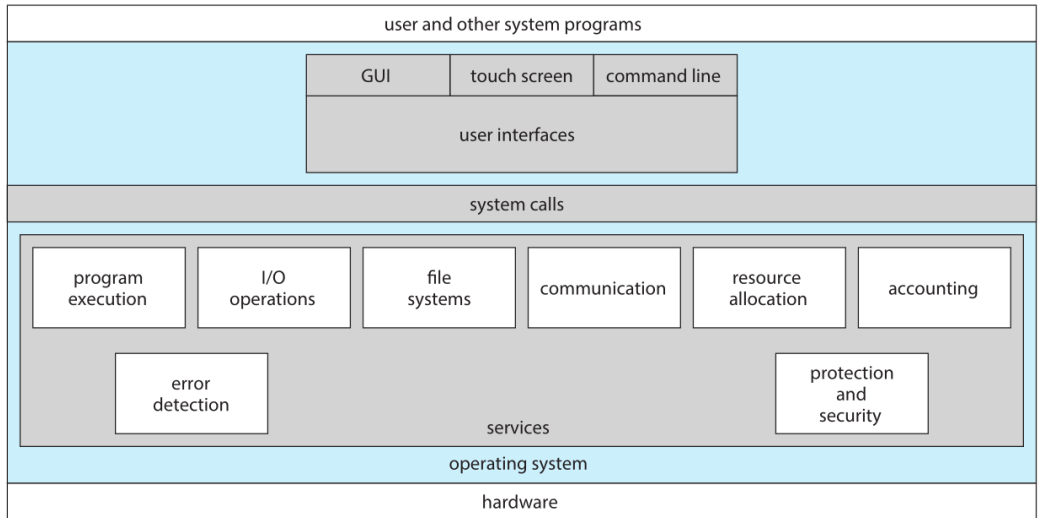


Figure: A view of operating system services.

System Calls

- **System Calls** – programming interface to services provided by the OS
- Usually written in **high-level languages** (C or C++)
- Programs normally access them through **APIs** rather than direct calls
 - Win32 API (Windows)
 - POSIX API (UNIX, Linux, macOS)
 - Java API (JVM)
- **Implementation**
 - Each system call has a unique **number**
 - System-call interface uses a table indexed by these numbers
 - OS kernel executes the requested service and returns results
- **Abstraction**
 - Programmer uses API without knowing internal implementation
 - Runtime libraries handle low-level details

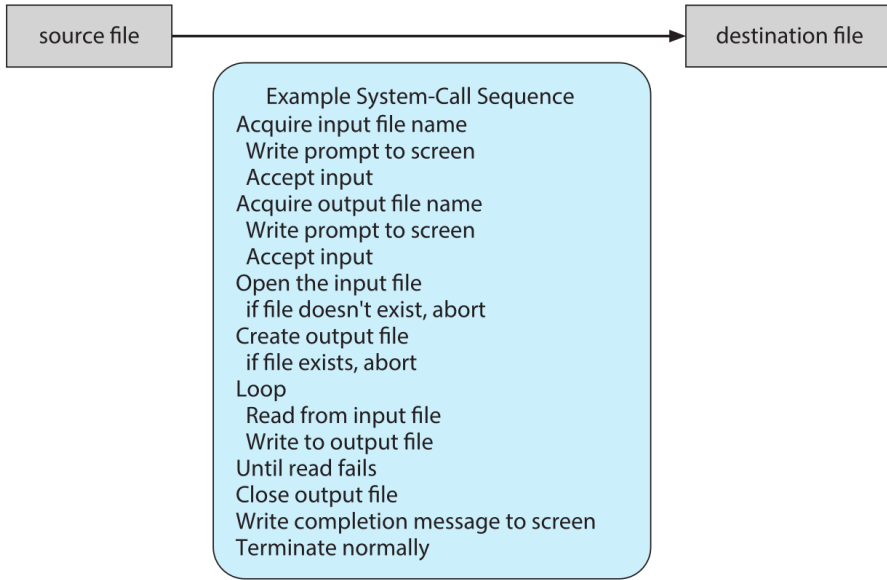


Figure: A view of operating system services.

Types of System Calls (1)

■ Process Control

- create process, terminate process
- end, abort
- load, execute
- get process attributes, set process attributes
- wait for time
- wait event, signal event
- allocate and free memory
- dump memory if error
- debugger support, single-step execution
- locks for managing shared data between processes

Types of System Calls (2)

■ File Management

- create file, delete file
- open, close file
- read, write, reposition
- get and set file attributes

■ Device Management

- request device, release device
- read, write, reposition
- get device attributes, set device attributes
- logically attach or detach devices

Types of System Calls (3)

■ Information Maintenance

- get time or date, set time or date
- get system data, set system data
- get and set process, file, or device attributes

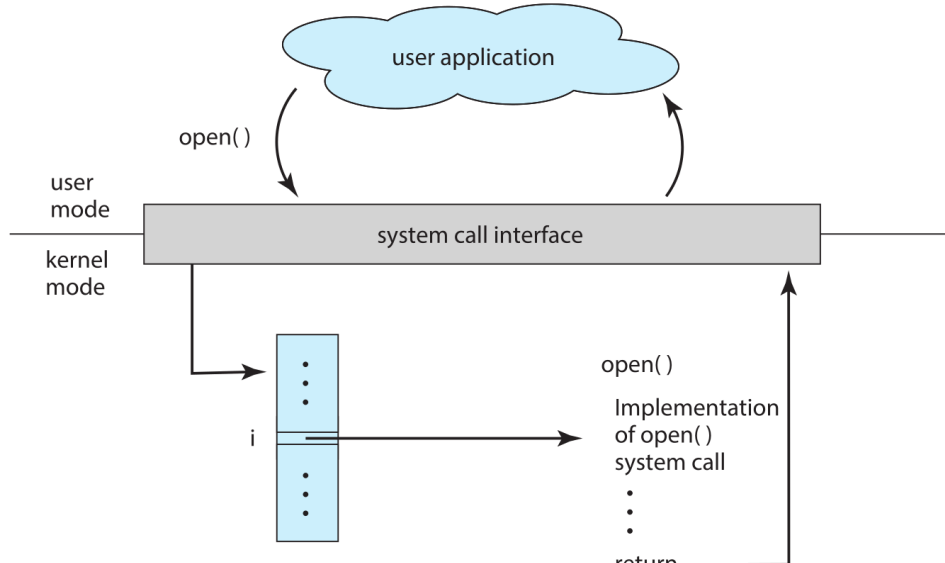
■ Communications

- create, delete communication connection
- send and receive messages (message passing)
- communication between client and server
- shared memory creation and access
- transfer status information
- attach and detach remote devices

■ Protection

- control access to resources
- get and set permissions
- allow or deny user access

API – System Call – OS Relationship



System Services

System Services (Overview)

- **System programs** provide a convenient environment for program development and execution
- They are typically grouped into:
 - **File manipulation**
 - **Status information**
 - **Programming language support**
 - **Program loading and execution**
 - **Communications**
 - **Background services**
 - **Application programs**
- For most users, the **view of the operating system** comes from system programs rather than direct system calls
- Some system programs act as simple **interfaces to system calls**, while others are more complex utilities

System Services (Examples)

■ File management

- Create, delete, copy, rename, print, list, and manipulate files and directories

■ Status information

- Provide system information such as date, time, memory, disk space, and users
- Offer performance monitoring, logging, and debugging information
- Some systems maintain a **registry** for configuration data

■ File modification

- Text editors for creating and editing files
- Tools for searching or transforming file contents

System Services (Development and Communication)

- **Programming language support**

- Compilers, assemblers, interpreters, and debuggers

- **Program loading and execution**

- Absolute loaders, relocatable loaders, linkage editors
- Overlay loaders and debugging systems

- **Communications**

- Mechanisms for communication between processes and systems
- Messaging between users, remote login, file transfer
- Web browsing and electronic mail

The Role of the Linker and Loader

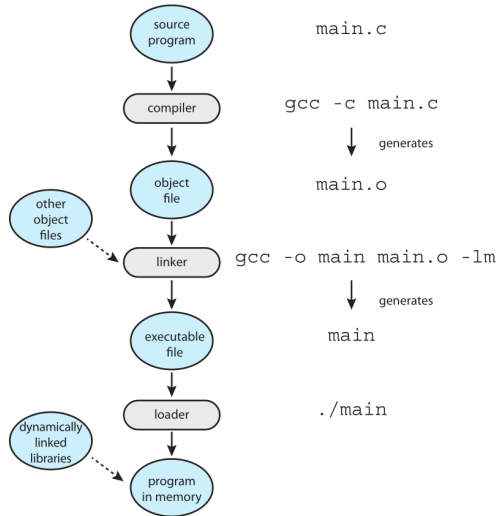


Figure: The role of the linker and loader.

Why Applications are Operating System Specific?

- Applications compiled for one OS usually **cannot run on another OS**
- Each operating system provides its own
 - **System calls**
 - **File formats** and system interfaces
- Applications can support **multiple operating systems**
 - Written in interpreted languages (Python, Ruby) with interpreters on different OSes
 - Written for a **Virtual Machine** (for example Java JVM)
 - Written in standard languages (like C) and compiled separately for each OS
- **Application Binary Interface (ABI)**
 - Binary-level equivalent of an API
 - Defines how compiled programs interact with the OS and hardware
 - Depends on architecture, CPU, and operating system

- 1 Avi Silberschatz, Peter B. Galvin, and Greg Gagne. *Operating System Concepts*. Global Edition. Wiley, 2023. ISBN: 9781119320913.
- 2 Avi Silberschatz, Peter Baer Galvin, and Greg Gagne. *Operating System Concepts, Tenth Edition: Lecture Slides*. Wiley, 2018. Available at:
<https://os-book.com/OS10/slide-dir/index.html>. Accessed: 2026-03-06.



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Course webpage: https://laltu-sardar.github.io/courses/corgos_2026.html.